Absorptive But Not Anechoic

Simulating an anechoic chamber before you trust it: what the models predict, which failures they catch, and what it costs to find out. Acoustic scope only.

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21 sources retrieved
9 paywalled · DOIs below
papers/anechoic-simulation/

The result everything else hangs off. In 2019 Anthony Nash qualified a 51 m^3 commercially built anechoic chamber against ISO 3745 Annex A. On pink noise in one-third-octave bands it **passed**; on pure tones at the same band centres it **failed** — a 2056 Hz tone fitted a decay of −26.4 dB per decade of distance, about 8 dB per doubling, which is not physically possible in a free field and is explicable only as a reflection cancelling the direct wave.

Why it matters here. Band noise averages over the interference pattern; a tone sits inside it, and drone noise is tonal. A chamber qualified with noise is not qualified for this rig — and a simulation excited with noise reproduces the passing answer while the room is failing.

In short, the chamber is absorptive but not anechoic.

A. Nash, *Qualification of an anechoic chamber*, ICA 2019 · on a chamber with foam wedges on all six surfaces, an expanded-metal floor grate, and no evidence its wedges were ever impedance-tube tested

"Simulating a chamber" is three jobs

NOT ONE

STAGE	PRODUCES	COST / CATCH
1 · Absorber unit cell FEM · Floquet · PML	α(f,θ) and complex surface impedance of the *real* wedge geometry	minutes · needs flow resistivity σ from an impedance tube
2 · Interior field image source, or FEM/BEM/FDTD	p(x, f) with phase preserved, along each traverse	minutes → days · locally-reacting shortcut; wave solvers scale as f^3
3 · Virtual qualification fit $b - 20\cdot\log(r/r_0)$	ΔLp vs the ISO tolerance table → a (frequency × distance × direction) envelope	free · must be driven with tones, not band noise

Only stage 2 is expensive, and the published methodology (Bonfiglio & Pompoli) avoids meshing the room at all: it turns the stage-1 impedance into a spherical-wave oblique-incidence reflection coefficient and predicts the field with complex image sources. That is why chamber design is tractable on a laptop.

Method stack

ORDERED BY COST

METHOD	VALID WHERE — AND THE CATCH
Equivalent fluid + TMM Delany–Bazley · Miki · JCA	α(f) of a flat layer from one parameter. Model choice moves the fitted σ by 2.5×
FEM unit cell	α(f,θ) of any wedge or pyramid profile. Needs σ, which needs a measurement
Complex image source	Interior field with phase, near-zero cost. Locally-reacting unless explicitly corrected
Room FEM / BEM / FDTD	The only route that models extended-reaction absorbers. 6–10 DOF per wavelength; doubling f_{max} costs 8× the DOF
Geometrical acoustics energy ray tracing	Carries energy, not phase → cannot represent cancellation → returns a clean, confident, wrong answer

What the simulations get wrong

The input data is the error, not the solver. Uncertainty work targets three parameters: absorber flow resistivity, absorber thickness, hard-wall absorption. Vorländer: the input quality needed for JND-accurate prediction isn't available from reverberation-room measurements.

The diagnostic tell. When two numerical methods agree with each other but both miss the measurement, the material data is wrong — not the solver.

A model can only fail the way you asked it to. The Nash result is not a modelling artefact; it is a question-selection artefact. Reproduce a published benchmark case before trusting a first run.

> **The load-bearing conclusion.** Nothing in this literature supports substituting a model for an ISO 26101-1 divergence-loss measurement. What the model buys is targeting — which frequencies, directions and distances are marginal, and therefore where the twenty measurements you can afford should go instead of the two hundred you cannot.

The workflow

STAGE A · DAYS, ON A LAPTOP

1. **Impedance tube** (ISO 10534-2) on one flat sample of the lining — the only physical measurement stage A needs.
2. **Fit σ with ≥3 models** and keep the spread as an uncertainty band instead of picking a winner.
3. **FEM the real geometry in a unit cell** — one wedge, Floquet periodicity, PML. Output α(f,θ) and impedance.
4. **Phase-preserving image-source model** of the room with those coefficients; receivers on all eleven arc directions; include untreated features explicitly.
5. **Run the virtual ISO 26101-1 test** per direction; output the frequency × distance × direction map.
6. **Excite it with tones** at ⅓-octave centres. Tone-only failure is the result, not an artefact.
7. **Then measure** the flagged directions. Stage B — room FEM below the crossover with extended-reaction elements — only if stage A is marginal.

With no chamber at all: time-gate, planned with the image-source method. Gate length t_g sets the lowest trustworthy frequency at ≈ $1/t_g$ (5.6 ms → 178 Hz); image-source geometry gives the mounting height that buys it — for a 5 ms floor bounce at 0.48 m spacing, ≥ 1.08 m above the floor. Run any gating experiment with `trigger.enabled = false`.

Twelve ways a chamber lies to you

ALL DOCUMENTED

01 **Cut-off is depth, not quality.** Roughly where wedge length = λ/4. A better material in the same depth barely moves it.

02 **Geometry carries the absorption.** ITMO: a flat disk of the lining measures α < 0.5 (290–1800 Hz); the pyramid of the same material reaches 0.996.

03 **Locally-reacting boundaries are wrong for porous wedges.** Real absorbers are extended-reaction and angle-dependent — the standard shortcut, the standard optimism.

04 **Your one input is ambiguous.** Same sample, five models: σ = 3437 / 5136 / 5994 / 6125 / 8701 Pa·s/m^2.

05 **Tones expose what noise hides.** The Nash result. Decisive for tonal rotor noise.

06 **A chamber is not anechoic — it is anechoic in an envelope.** ITMO qualifies 50–3150 Hz at 0.5–0.9 m in every direction; past that the four directions diverge. Worst deviation: 15.5 dB.

07 **Grates, ports, doors and recesses are the actual reflectors.** ISO 26101-1 requires traverses aimed at exactly these features.

08 **Your own rig is inside the field.** Booms, arc frame, turntable — scatter that reads as directivity after the fact.

09 **The fit changes the verdict.** ISO 3745 fits $a/(r-r_0)$ with a fitted acoustic-centre offset; ISO 26101 fits $b - 20\cdot\log(r/r_0)$, r_0 = 1 m. Not always the same answer.

10 **Small rooms can be unqualifiable by rule, not physics.** Traverse minimum was λ/2 in ISO 26101:2012, λ/4 since 2017. Split's 2.8 × 1.7 × 2.05 m chamber bottomed out at 250 Hz.

11 **Nobody ever tested the wedges.** A purchased chamber is not a qualified one; one qualified once is not one qualified now.

12 **A ray-tracing model will say it's fine.** Brinkmann's 2019 round robin: obvious model errors "once the assumptions of geometrical acoustics are no longer met"; no algorithm matched even a single reflection's comb-filter structure; low-frequency reverberation over by 58 % at 125 Hz.

What this means for the arc rig

ISO 5305:2024

ISO 5305 (rotor UAS < 150 kg) does not say "use a chamber". It requires ISO 26101-1 inverse-square validity *for all source positions and measurement directions specified*, with traverses *along the measurement directions defined by the microphone positions.* For an 11-mic arc at 18° that is up to eleven traverse directions, not five.

Tolerance	±1.5 / ±1.0 / ±1.5 dB	125–630 · 800–5000 · ≥6300 Hz
Far field	$R \geq 5\cdot D_A$	0.64 m for a 5″ prop; 0.76 m for 6″
Ground clearance	$H_0 \geq \max(2\cdot D_A,\ 1.2\ \text{m})$	aerodynamic rule; also sets floor-bounce path
Background noise	≥ 6 dB, prefer ≥ 15	failing bands must be stated, not silently included
Mic supports	acoustically treated	the arc frame is a scatterer in the field

> **The arc makes the room error worse, not neutral.** Chamber deviation is direction-dependent. A single microphone samples one direction and picks up an offset; an elevation arc samples eleven simultaneously and plots them as a polar shape. A direction-dependent room artefact therefore lands on the *shape* of the directivity plot and looks exactly like real directivity — the same structural hazard as the recirculation finding, moved from the time axis to the spatial one. Comparisons at a *fixed* elevation stay far better protected, because the error is common-mode there.

Needed before stage 2 can run

BLOCKING

- **The arc radius.** Not in `config.toml`, not in `docs/mic_arc.md` (which has all 11 serials and elevations). It decides whether the 275 stored takes are far-field at all.
- **Room dimensions and surfaces** — L × W × H, floor material, wall treatment, where the rig sits. The entire geometric input to stage A step 4.
- **Rotor-plane height** above the floor, and whether it can move. **Prop diameters** in the dp1 campaign, for the far-field and clearance numbers.
- **Is a chamber actually on the table?** Budget and space — or is this about characterising the room you already measure in? Stage A is identical either way; only what you do with the answer changes.

FULL REPORT & PRIMARY EVIDENCE

Full review — interactive, with figures
claude.ai/code/artifact/6226a13c-4e3e-43e6-bf94-a94fc8ca c625

Nash — Qualification of an anechoic chamber, ICA 2019
pub.dega-akustik.de/ICA2019/data/articles/001379.pdf

Bikmukhametov et al. — ITMO qualification envelope
arxiv.org/abs/2603.16556

STANDARDS & QUALIFICATION METHOD

ISO 26101-1:2021 — free-field qualification (preview)
cdn.standards.iteh.ai · ISO-26101-1-2021.pdf

ISO 5305:2024 — UAS noise measurement (preview)
cdn.standards.iteh.ai · ISO-5305-2024.pdf

Russo et al. — ISO 3745 vs ISO 26101, Euronoise 2018
euronoise2018.eu/docs/papers/369_Euronoise2018.pdf

FULL TEXTS NOW HELD — RE-CHECKED 2026-09-07

All eight core papers (Bonfiglio & Pompoli, Schneider, Jiang, Cunefare, Singh, Wang & Tang, Vorländer, Brinkmann) are in `papers/anechoic-simulation/` and `papers/chamber-problems/`; every abstract-tier claim in the parent review was re-checked against them and the round-robin figures on this page corrected. Still unread: Easwaran & Munjal 1993, Kar & Munjal 2006, Tavakkoli Nejad 2020. Companion brief: `docs/anechoic-chamber-problems-brief.pdf`.

Also read at source · Beranek, Sleeper & Moots 1945 (OSRD 4190) govinfo.gov · GOVPUB-PR32_413 · Prinn, FEM for room acoustics mdpi.com/2624-599X/5/2/22 · González & Kob, DAGA 2026 dega-akustik.de · DAGA_2026 688.pdf · ARTA AN4 time-gating artalabs.hr · AN4-FreeField · github.com/LCAV/pyroomacoustics · full source list and figures in the interactive report above